

Shenzhen Dudumiao Technology Co., Ltd.

smokefire Fire, Smoke & Smoking Detection

Deployment Guide

Document version: V1.0

Applicable software version: 1.0.0

Updated: 2026-08-03

Contents

Document information	2
1. Delivery method and requirements	2
1.1 Standard bundle	2
1.2 Host requirements	2
2. Download and verification	3
3. Windows deployment	3
4. Linux deployment	4
5. First start and acceptance	5
6. Camera preparation	5
7. Configuration and LAN access	6
8. Routine operation	6
9. Backup, restore, and upgrade	7
10. Troubleshooting	7
11. Handover checklist	8
Installation	8
Business acceptance	8

Important

smokefire is an AI-assisted video alerting tool, not a certified fire alarm device. It does not replace statutory fire protection, human patrols, emergency procedures, or on-site verification. False alerts and missed events are possible.

Document information

Item	Value
Document	smokefire Deployment Guide
Audience	Implementation engineers, administrators, and site operators
Standard delivery	Standalone Docker CPU bundle without the project source tree
Host platforms	Windows 10/11, Windows Server, mainstream x86_64 Linux
Default URL	http://127.0.0.1:8600
Project	https://github.com/newtv-ai/smokefire

1. Delivery method and requirements

1.1 Standard bundle

Download [smokefire-deploy-1.0.0-cpu.zip](#) from GitHub Release [v1.0.0](#). The bundle contains:

- a prebuilt smokefire CPU Docker image;
- [fire_smoke_v5.pt](#) and [smoking_v4.pt](#) model weights;
- Docker Compose configuration;
- Windows and Linux start/stop scripts;
- Chinese and English deployment, operation, and model-test PDFs;
- a SHA-256 manifest.

The bundle does not contain the project source tree. The target host does not need Python, Git, or build tools, and the service does not require GitHub after installation.

1.2 Host requirements

Item	Recommended minimum	Notes
CPU	x86_64, 4 cores	Benchmark the planned stream count; the CPU package targets pilots and smaller installations
Memory	8 GB	16 GB or more is recommended
Disk	20 GB free	Add capacity for event video, snapshots, and backups
Docker	Docker Engine/Desktop 24+	Docker Compose v2 is required

Item	Recommended minimum	Notes
Browser	Maintained Chrome or Edge	Used for the web interface
Camera	RTSP, H.264 recommended	The host must reach each camera or NVR

The standard package is CPU-only. Multi-stream production use requires validation with actual streams on the target host. Obtain a separate GPU bundle matched to the NVIDIA driver and CUDA environment when GPU acceleration is required.

2. Download and verification

Open <https://github.com/newtv-ai/smokefire>, go to Releases, select **v1.0.0**, and download:

```
smokefire-deploy-1.0.0-cpu.zip
```

On Windows, extract to a path without non-ASCII characters or spaces, for example **D:\smokefire**.

Linux example:

```
sudo mkdir -p /opt/smokefire
sudo unzip smokefire-deploy-1.0.0-cpu.zip -d /opt/smokefire
cd /opt/smokefire
```

Verify on Windows:

```
powershell -ExecutionPolicy Bypass -File .\verify.ps1
```

Verify on Linux:

```
chmod +x verify.sh start.sh stop.sh
./verify.sh
```

Model reference hashes:

File	Size	SHA-256
models/fire_smoke_v5.pt	44,014,233 bytes	b5248d55c5d90e341bc4e537887d4bce4b9af1bd0330ddd87825294750a1e216
models/smoking_v4.pt	44,025,689 bytes	0ea7b66c2105f1898f56f828b18ffb01934fb1f47a795192e8867ae8fcb128dd

Do not continue when verification fails. Download the affected file again.

3. Windows deployment

Install Docker Desktop, select Linux containers, and wait for the engine to start. In PowerShell, verify:

```
docker version
docker compose version
```

From the extracted bundle directory, start smokefire:

```
powershell -ExecutionPolicy Bypass -File .\start.ps1
```

The script verifies the package, imports the prebuilt image, starts the container, and waits for the health check. The first image import can take several minutes.

Status and logs:

```
docker compose ps
docker compose logs -f smokefire
```

Ctrl+C exits log viewing without stopping the service.

Stop and restart:

```
powershell -ExecutionPolicy Bypass -File .\stop.ps1
powershell -ExecutionPolicy Bypass -File .\start.ps1
```

The stop script preserves the event-data volume.

4. Linux deployment

Install Docker Engine and Compose v2 using the distribution's official instructions, then verify:

```
docker version
docker compose version
```

Add **sudo** if the current account does not have Docker access, or have an administrator configure the Docker group and log in again.

Start:

```
cd /opt/smokefire
chmod +x verify.sh start.sh stop.sh
./start.sh
```

Status and logs:

```
docker compose ps
docker compose logs -f smokefire
```

Stop and restart:

```
./stop.sh
./start.sh
```

`restart: unless-stopped` brings the container back when Docker restarts. Validate one complete host reboot before production handover.

5. First start and acceptance

Open <http://127.0.0.1:8600> on the service host. Loading both models for the first time may take from tens of seconds to several minutes.

Health endpoints:

Endpoint	Purpose	Expected result
/api/health/live	Process liveness	HTTP 200
/api/health/ready	Models, scheduling, recording, and disk readiness	HTTP 200
/api/status	Runtime and notification metrics	JSON response

```
curl http://127.0.0.1:8600/api/health/live
curl http://127.0.0.1:8600/api/health/ready
```

Minimum acceptance:

- 1 Open Live Wall, Event Center, and Camera Management.
- 2 Add one authorized RTSP camera.
- 3 Confirm the camera becomes Online and the image/name are correct.
- 4 Trigger one alert with authorized test material or a controlled site test.
- 5 Confirm the snapshot, class, confidence, time, and video in Event Center.
- 6 Test both Confirmed and False Positive actions.
- 7 Restart the service and confirm camera configuration and history remain.

Successful installation is not final production acceptance. Validate capacity, latency, false alerts, misses, evidence completeness, stream recovery, and continuous operation with the real network, storage, and planned camera count.

6. Camera preparation

Prepare the camera name, RTSP URL, credentials, NVR subnet, codec, resolution, bitrate, and planned stream count.

Windows connectivity check:

```
Test-NetConnection <camera-ip> -Port 554
```

Linux:

```
nc -vz <camera-ip> 554
```

Example RTSP URL:

```
rtsp://user:password@192.168.1.20:554/stream1
```

Prefer H.264 main streams or a substream sized for available compute. Validate H.265, very high resolutions/bitrates, and vendor-specific packaging separately.

7. Configuration and LAN access

The bundle binds the web port to `127.0.0.1:8600` by default. For access from other LAN computers, the implementer must bind the port to the intended private IP in `docker-compose.yml`, set the allowed host names in `.env`, and restart the service.

Common `.env` settings:

Setting	Default	Purpose
SMOKEFIRE_ALLOWED_HOSTS	localhost,127.0.0.1	Allowed HTTP Host values
SMOKEFIRE_WEBHOOK_URL	empty	Global webhook endpoint
SMOKEFIRE_INFER_WORKERS	4	CPU inference workers
SMOKEFIRE_INFER_BATCH_MAX	8	Maximum inference batch
SMOKEFIRE_STOP_GRACE_PERIOD	180s	Graceful shutdown allowance

Restart after global environment changes. Per-camera detection settings apply after saving in Camera Management.

Camera credentials may appear inside RTSP URLs. Restrict access to the deployment directory, backups, and terminal history, and never expose full credentials in screenshots or tickets.

8. Routine operation

Recommended daily checks:

```
docker compose ps
docker compose logs --tail 200 smokefire
curl http://127.0.0.1:8600/api/health/ready
```

Also confirm that cameras are online, previews update, recent events play, and disk usage is below the high-water threshold. Do not run `docker compose down -v`; it deletes the named business-data volume.

9. Backup, restore, and upgrade

Business data resides in the `smokefire-data` named volume. Stop the service before backup when practical, record the current version, and use an administrator-approved Docker-volume method to copy the complete `/app/data` tree.

A complete backup includes camera configuration, the SQLite database, event clips, snapshots, false-positive feedback, and the notification queue. A database-only copy is not a complete evidence backup.

Restore:

- 1 Stop the service.
- 2 Restore the complete data tree for the same software version.
- 3 Start the service.
- 4 Verify health, cameras, history, clips, and feedback.

Upgrade:

- 1 Back up the data volume and confirm the backup is readable.
- 2 Download and verify the new bundle.
- 3 Stop the old version and run the new bundle's start script.
- 4 Repeat health and minimum business acceptance.
- 5 If required, stop the new version and restore the old image plus the pre-upgrade backup.

10. Troubleshooting

Symptom	Check first	Action
Docker command unavailable	Docker Desktop/Engine state	Start the engine and retry <code>docker version</code>
Image archive missing	Completeness of <code>images/</code> and the ZIP	Download again and run verification
Readiness does not pass	Container logs, model hashes, free disk	<code>docker compose logs --tail 300 smokefire</code>
Web page unavailable	Container state, port 8600, bind address	Test <code>127.0.0.1</code> on the host first
Camera keeps reconnecting	RTSP URL, credentials, port 554, codec	Test the same RTSP URL in a player on the host
Preview black while Online	Browser decode, jitter, first closed segment	Refresh and inspect stream/FFmpeg logs
No alerts	Capability switch, thresholds, target size	Use an authorized positive sample; do not sharply lower thresholds without replay testing
Too many alerts	Steam, fog, reflections, eating, handheld objects	Collect false positives and tune against the same positive/negative set
Event has no clip	Disk, recording warm-up, source stability	Check storage and FFmpeg logs
Disk nearly full	Retention and backup policy	Back up first, then delete by date in Storage Management

11. Handover checklist

Installation

- ☐ Bundle, image, and both model hashes verified.
- ☐ Start, stop, restart, and host-reboot recovery tested.
- ☐ `/api/health/live` and `/api/health/ready` healthy.
- ☐ All three business pages open.
- ☐ Installation directory, data volume, backup location, and version recorded.
- ☐ Backup and restore method tested.

Business acceptance

- ☐ Planned cameras connected with correct names, time, and images.
- ☐ Alert snapshots and event clips play correctly.
- ☐ Confirmed, false-positive archive, and feedback export tested.
- ☐ Webhook delivery tested when enabled.
- ☐ Target-scene positives and negatives used to validate misses, false alerts, and alert latency.
- ☐ Capacity, reconnect behavior, and continuous operation tested at the planned stream count.
- ☐ Users received the Operation Manual and Model Capability Test Guide and completed handover training.

Prepared by: Shenzhen Dudumiao Technology Co., Ltd. Project:
<https://github.com/newtv-ai/smokefire>